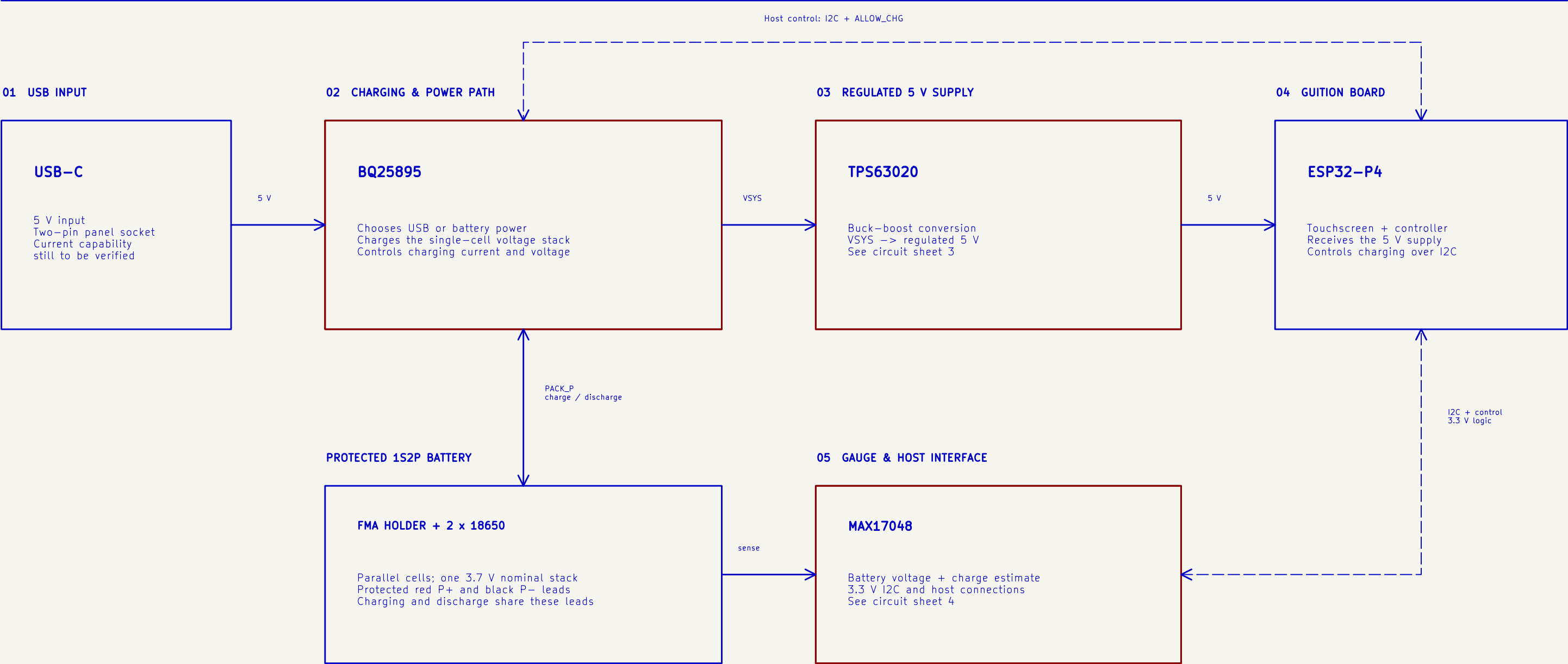


Power system overview

Read from left to right. Double-click a numbered circuit sheet, or select it in the Hierarchy Navigator and press Enter.



06 HOW TO READ THE CIRCUIT SHEETS

- Each outlined section performs one job. Read its title before tracing individual wires.
 - Matching net labels are connected electrically, including across different sheets.
 - Component prefixes: U = IC, R = resistor, C = capacitor, L = inductor, J = connector.
 - A dot joins crossing wires. A cross on a pin means deliberately unconnected.
 - DNP means the component is drawn but not fitted in the initial assembly.
- Overview: solid arrows show power/sensing; dashed arrows show host communication.
These arrows show function; the detailed sheets contain the actual wiring.

07 DESIGN STATUS

Review draft; not released for fabrication.
USB input, cells and final parts need verification.
Charging currently requires an active host.

Trimix

Sheet: /
File: Trimix_Power.kicad_sch

Title: Trimix power system – replacement schematic

Size: A3 Date: 2026-09-05

KiCad E.D.A. 10.0.6

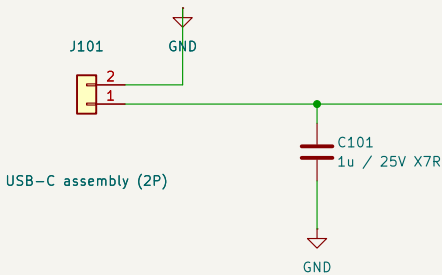
Rev: P1.1

Id: 1/4

Charging & protected battery

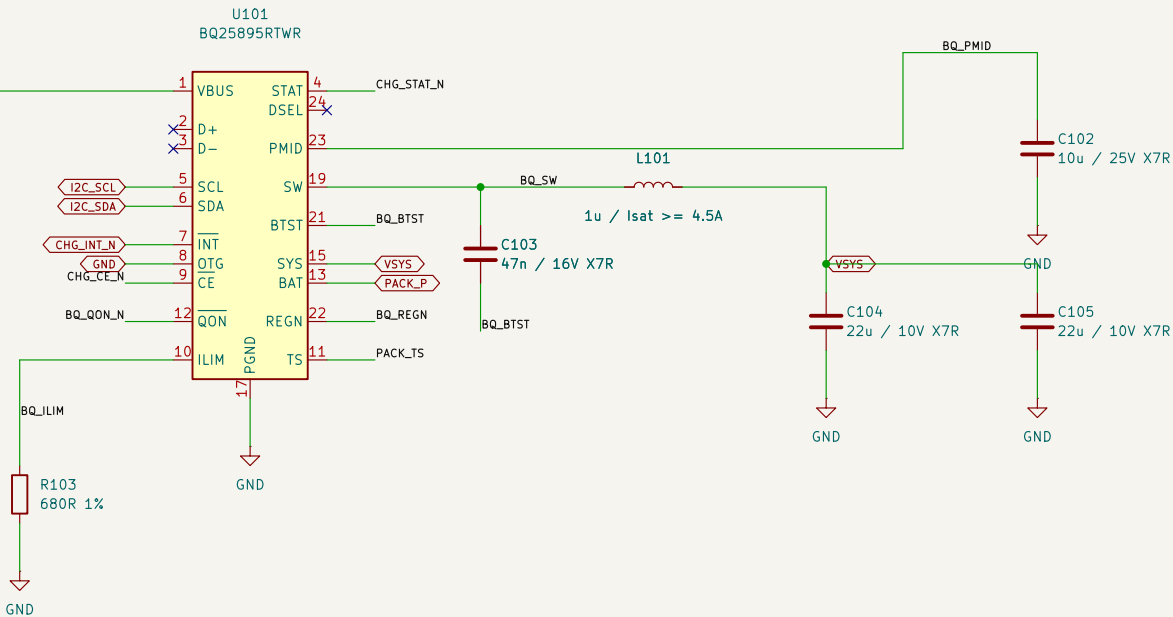
Follow the top row from USB input to charger and battery. The lower row controls and monitors charging.

01 USB INPUT

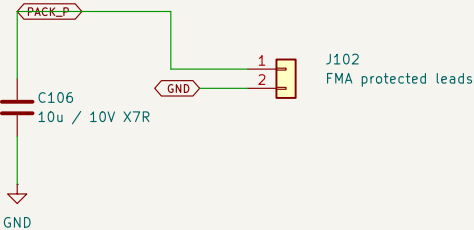


J101 brings 5 V from the two-pin USB-C assembly. C101 bypasses the charger input. CC termination and available current remain unverified.

02 CHARGER & POWER PATH

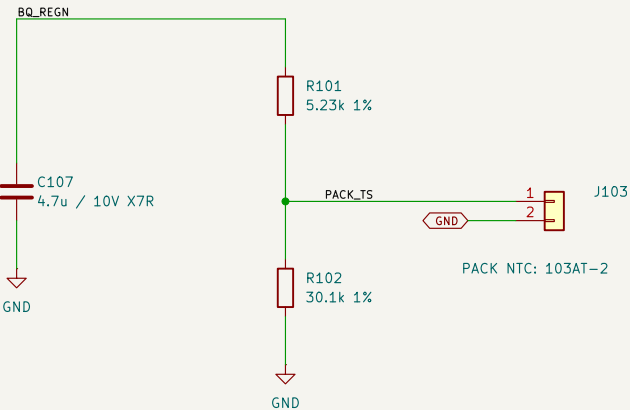


03 PROTECTED BATTERY



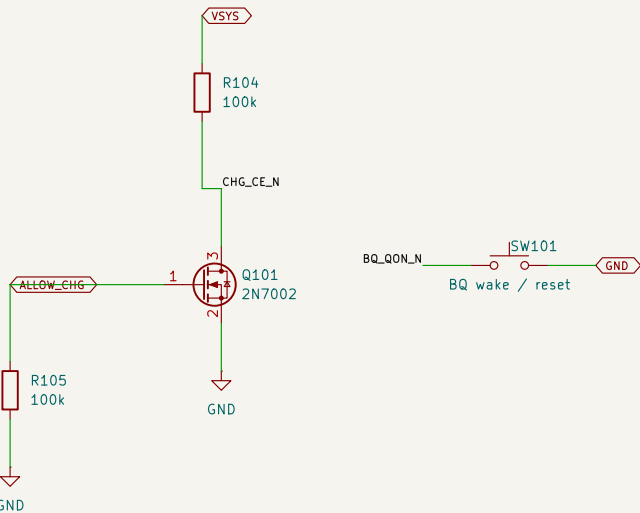
P+ = protected red lead. P- = protected black lead. Both 18650 cells sit in parallel in the FMA holder. The holder supplies the battery protection. PCB ground always connects to protected P-.

04 PACK TEMPERATURE



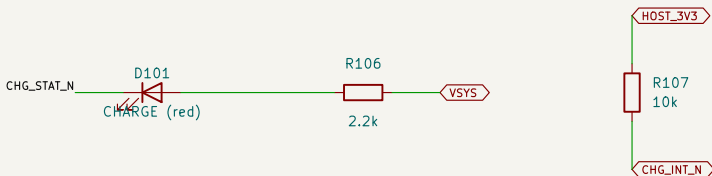
R101/R102 and the external 10k NTC sense temperature. Attach the NTC to the cells; connect it at J103. An open sensor inhibits charging.

05 CHARGE ENABLE & WAKE



HIGH at ALLOW_CHG turns Q101 on and enables charging. R105 holds charging off while the host resets. SW101 provides charger wake/reset, not a power switch.

06 STATUS INDICATORS



D101 lights while the charger reports charging. R106 limits LED current. R107 pulls the interrupt up to the host 3.3 V rail.

IINLIM: 450 mA / ICHG: 512 mA / VREG: 4.192 V
These are draft firmware targets, not factory defaults. Configure and verify the charger before ALLOW_CHG.

READING THIS SHEET

Matching labels connect, even when no wire is drawn between them. VSYS feeds the 5 V converter on sheet 3. Crossed pins are intentionally unused. Check input limits, cell ratings and capacitor derating before hardware tests. Component selection and firmware requirements are recorded in hardware/POWER_DESIGN.md.

Review draft | Electrical design unchanged | See POWER_DESIGN.md

Trimix

Sheet: /Charging and protected pack/

File: Charging.kicad_sch

Title: USB charging and protected 1S2P pack

Size: A3

Date: 2026-09-05

Rev: P1.1

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Id: 2/4

01 INPUT FILTERING

The schematic diagram illustrates the input filter network for the TPS7A01. It features a common-mode choke (C201) connected to the VSYS input and ground (GND). A series capacitor (C202) is connected between the choke and the input of the TPS7A01. A shunt capacitor (C203) is connected between the output of the series capacitor and ground (GND). The output of the filter is labeled TPS_VINA.

C204–C206 support load steps and reduce output ripple.

The diagram shows a 5V output line (VOUT_5V) connected to three parallel capacitors: C204, C205, and C206. Each capacitor is specified as 22u / 10V X7R and is connected to ground (GND). This configuration is used to support load steps and reduce output ripple.

VSYS

R203
100k

POWER_EN

GND

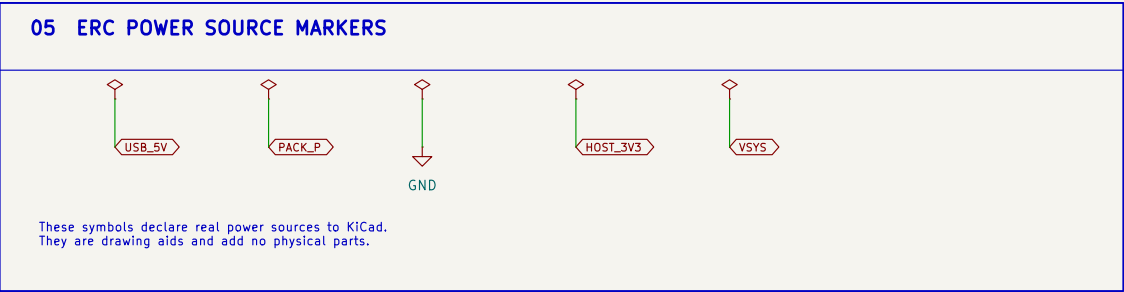
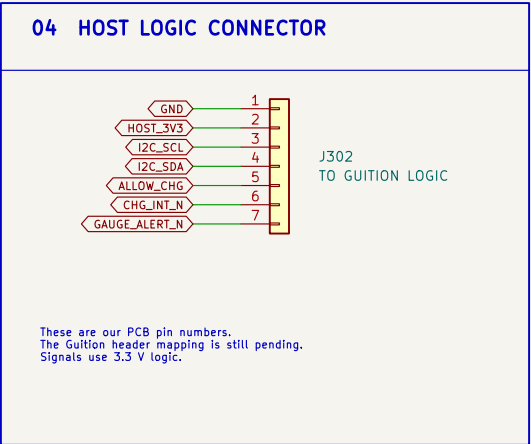
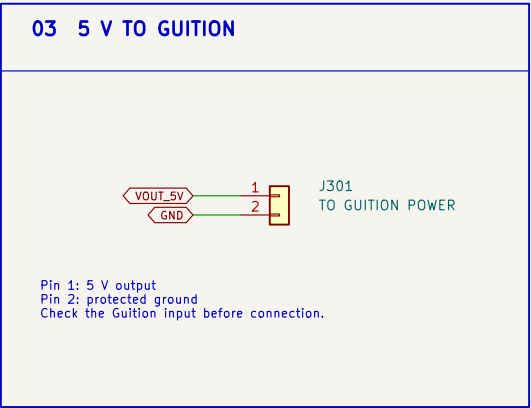
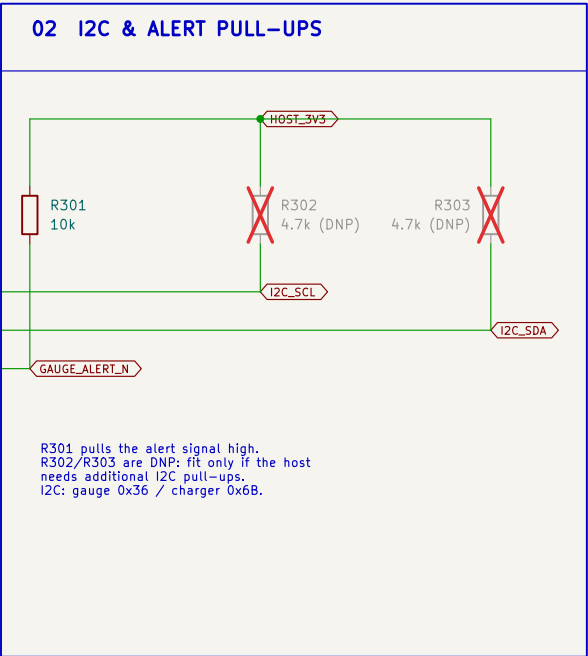
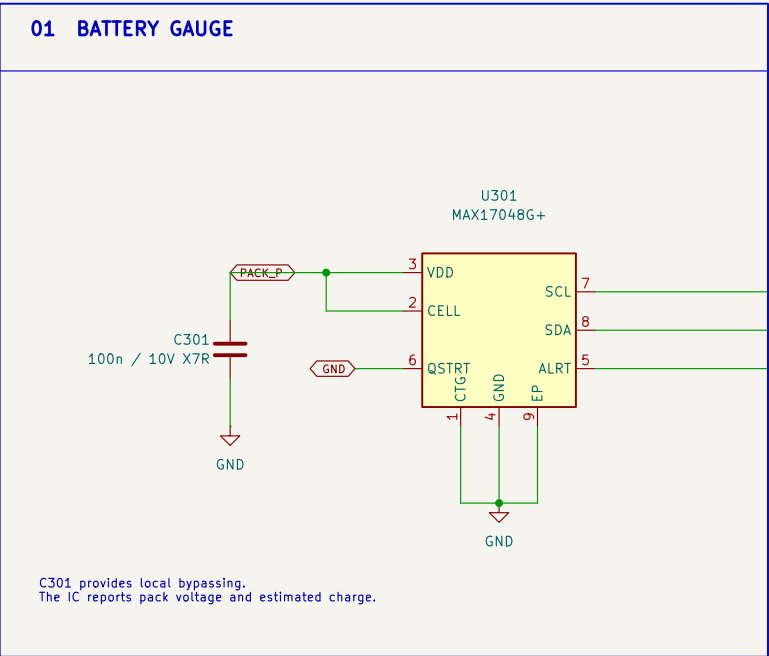
J201
OFF: short EN to GND

5.000 V = 0.5 V x (1 + 1.62M / 180k)
R201/R202 feed a fraction of the output back to FB.
Use 1% resistors; confirm the output on the bench.

Id: 3/4

Battery gauge & host interface

The fuel gauge reads the protected pack. All communication pull-ups use the Guition board's 3.3 V supply.



PACK_P = protected red P+ / GND = protected black P-.
DNP = do not populate this component in the initial assembly.

Review draft | Electrical design unchanged | See POWER_DESIGN.md

Trimix

Sheet: /Fuel gauge and Guition interface/

File: Gauge_Interface.kicad_sch

Title: Battery gauge and Guition interface

Size: A4

Date: 2026-09-05

Rev: P1.1

KiCad E.D.A. 10.0.6

Id: 4/4